

第一题: (20 分)

- 1) 不考虑相对论, 电子经过高电压 V 加速, 求德布罗意波长;
- 2) 写出坐标表象下 x 和 p 的表达式, 求其对易关系 $[x, p]$;
- 3) 一维体系, 当势函数为偶函数即 $V(x) = V(-x)$ 时, 求证波函数总有确定的宇称;

$$\begin{aligned} (1) \quad \lambda &= \frac{h}{p} & qU &= \frac{1}{2}mv^2 \\ &= \frac{h}{mv} & v &= \sqrt{\frac{2qU}{m}} \\ &= \frac{h}{\sqrt{2mqU}} \end{aligned}$$

$$\begin{aligned} (2) \quad \hat{x} &= x \\ \hat{p} &= -i\hbar \frac{\partial}{\partial x} \\ [\hat{x}, \hat{p}] &= i\hbar \end{aligned}$$

$$\begin{aligned} (3) \quad &-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) + V(x) \psi(x) = E \psi(x) \\ &-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(-x) + V(x) \psi(-x) = E \psi(-x) \\ &\quad \text{令 } -x = x' \\ &-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \psi(x') + V(-x') \psi(x') = E \psi(x') \\ &\therefore \psi(x') \text{ 即 } \psi(-x) \text{ 也是同一薛定谔方程的解} \\ &\therefore \psi(x) \pm \psi(-x) \text{ 也是同一方程的解} \\ &\quad \text{波函数有确定的宇称} \end{aligned}$$

第二题: (30 分)

一无限深方势阱两边坐标为 $-a$ 和 a , 存在一 δ 势垒, $V(x) = aV_0\delta(x)$ 试

根据以下步骤求解:

- 1) 写出薛定谔方程, 及波函数的表达式
- 2) 写出波函数在 $\pm a$ 处的边界条件以及在 $x=0$ 处的跳跃条件;
- 3) 势能具有对称性, 故波函数有确定的宇称。试求奇宇称的波函数及其能级, 与无限深方势阱的能级对比;
- 4) 求偶宇称解的能级满足的公式, 通过画图法求解本征能量, 并讨论当 a 很大或者很小的情况, 对比奇宇称能级, 说明哪一个

是基态。

$$(1) \quad -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E \psi(x) \quad |x| < a \quad \text{且 } x \neq 0$$

$$\psi = \sqrt{\frac{1}{a}} \sin \frac{n\pi}{2a} (x+a)$$

$$(2) \quad \psi'(0^+) - \psi'(0^-) = \frac{2maV_0}{\hbar^2} \psi(0)$$

$$\psi(\pm a) = 0$$

(3) 奇宇称

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\psi(x) = A \sin kx + C \cos kx \quad 0 < x < a$$

$$\psi(x) = B \sin kx + D \cos kx \quad -a < x < 0$$

$$\psi(x) = -\psi(-x) \quad \psi(a) = 0$$

$$C = D = 0$$

$$ka = n\pi$$

$$k = \frac{n\pi}{a}, \quad n = 1, 2, 3, \dots$$

$$A = B$$

$$A^2 \int_{-a}^a \sin^2 kx \, dx = 1$$

$$\sqrt{\frac{2mE}{\hbar^2}} = \frac{n\pi}{a}$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2ma^2} = \frac{n^2 \pi^2 \hbar^2}{8ma^2}, \quad n = 2, 4, 6, 8, \dots$$

$$A^2 \cdot \int_{-a}^a \frac{1 - \cos 2kx}{2} \, dx = 1$$

$$A^2 \cdot \left(\frac{1}{2}x - \frac{\sin 2kx}{4k} \right) \Big|_{-a}^a = 1$$

$$A = \sqrt{\frac{1}{a}}$$

$$\psi = \sqrt{\frac{1}{a}} \sin \frac{n\pi x}{a} \quad n = 2, 4, 6, 8, \dots$$

(3) 偶宇称

$$\psi(x) = \psi(-x)$$

$$\psi(x) = A \sin kx + B \cos kx \quad 0 < x < a$$

$$\psi(x) = C \sin kx + D \cos kx \quad -a < x < 0$$

$$\psi(x) = \psi(-x), \quad \psi(a) = \psi(-a) = 0, \quad \psi'(0^+) - \psi'(0^-) = \frac{2maV_0}{\hbar^2} \psi(0)$$

$$B = D$$

$$A = -C$$

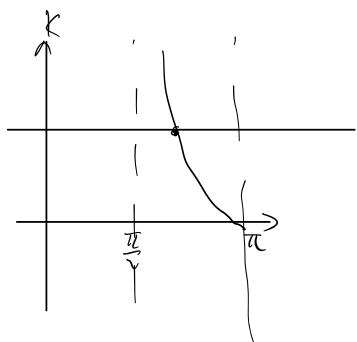
$$\psi(x) = -C \sin kx + B \cos kx \quad (0 < x < a)$$

$$\psi(x) = C \sin kx + B \cos kx \quad (-a < x < 0)$$

$$-C \sin ka + B \cos ka = 0$$

$$B = \tan ka \cdot C \quad (1)$$

$$-kC - kC = \frac{2maV_0}{\hbar^2} B$$



$$-2kc = \frac{2maV_0}{\hbar^2} B \quad \text{②}$$

$$\frac{\text{①}}{\text{②}} \text{ 得 } \frac{\hbar^2}{2maV_0} = -\frac{\tan ka}{2k}$$

$$k = -\frac{maV_0 \tan ka}{\hbar^2}$$

第三题: (25 分)

一维谐振子, 势能函数为 $V(x) = \frac{1}{2}m\omega^2 x^2$, 升降算符为

$a_{\pm} = \frac{1}{\sqrt{2\hbar m\omega}}(\mp ip + m\omega x)$, 且已知 $a_+ \psi_n = \sqrt{n+1} \psi_{n+1}$, $a_- \psi_n =$

$\sqrt{n} \psi_{n-1}$, 充分利用升降算符的性质, 试求

- 1) 若粒子处于量子态 ψ_n , 求位置和动量的平均值 $\langle x \rangle, \langle p \rangle$;
- 2) 若粒子处于量子态 $\psi_n, \langle x^2 \rangle, \langle p^2 \rangle$;
- 3) 求不确定度 $\sigma_x = \langle (x - \bar{x})^2 \rangle$, $\sigma_p = \langle (p - \bar{p})^2 \rangle$, 并验证不确定度关系, 说明等号什么情况下成立;
- 4) 若粒子处于 $\psi = a\psi_1 + b\psi_2$ 量子态, 求位置的平均值;
- 5) 上问中波函数经过 t 时间的演化后, 求位置的期望。

$$(1) \quad \langle x \rangle = \int \psi_n^* x \psi_n dx = 0$$

$$x \psi_n = \frac{1}{\alpha} \left(\sqrt{\frac{n}{2}} \psi_{n-1} + \sqrt{\frac{n+1}{2}} \psi_{n+1} \right)$$

$$\langle p \rangle = \int \psi_n^* \left(-i\hbar \frac{\partial}{\partial x} \right) \psi_n dx = 0$$

$$(2) \quad \langle x^2 \rangle = \int_{-\infty}^{\infty} (x \psi_n)^* x \psi_n dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\alpha^2} \left(\sqrt{\frac{n}{2}} \psi_{n-1}^* + \sqrt{\frac{n+1}{2}} \psi_{n+1}^* \right) \left(\sqrt{\frac{n}{2}} \psi_{n-1} + \sqrt{\frac{n+1}{2}} \psi_{n+1} \right) dx$$

$$= \frac{1}{\alpha^2} \left(\frac{2n+1}{2} \right) = \frac{1}{\alpha^2} \left(n + \frac{1}{2} \right)$$

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} \alpha^2 \left(\sqrt{\frac{n}{2}} \psi_{n-1}^* - \sqrt{\frac{n+1}{2}} \psi_{n+1}^* \right) \left(\sqrt{\frac{n}{2}} \psi_{n-1} - \sqrt{\frac{n+1}{2}} \psi_{n+1} \right) dx$$

$$= \hbar^2 \alpha^2 \left(n + \frac{1}{2} \right)$$

$$(3) \quad \sigma_x = \sqrt{\frac{1}{\alpha^2} \left(\frac{2n+1}{2} \right)}$$

$$\sigma_p = \hbar \sqrt{\alpha^2 \left(n + \frac{1}{2} \right)}$$

$$\sigma_x \cdot \sigma_p = \hbar \left(n + \frac{1}{2}\right) \geq \frac{\hbar}{2}, \quad n = 0, 1, 2, \dots$$

$$\begin{aligned} (4) \quad \langle x \rangle &= \int (a^* \psi_1^* + b^* \psi_2^*) x (a \psi_1 + b \psi_2) dx \\ &= \int (a^* b \psi_1^* x \psi_2 + b^* a \psi_2^* x \psi_1) dx \\ &= \frac{2ab}{\alpha} \end{aligned}$$

$$(5) \quad |\psi(t)\rangle = a \psi_1 e^{-\frac{iE_1 t}{\hbar}} + b \psi_2 e^{-\frac{iE_2 t}{\hbar}}$$

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{+\infty} (a^* \psi_1^* e^{\frac{iE_1 t}{\hbar}} + b^* \psi_2^* e^{\frac{iE_2 t}{\hbar}}) x (a \psi_1 e^{-\frac{iE_1 t}{\hbar}} + b \psi_2 e^{-\frac{iE_2 t}{\hbar}}) dx \\ &= \int_{-\infty}^{+\infty} [a^* b \psi_1^* x \psi_2 e^{\frac{i}{\hbar}(E_1 - E_2)t} + b^* a \psi_2^* x \psi_1 e^{\frac{i}{\hbar}(E_2 - E_1)t}] dx \\ E_1 &= \frac{3}{2} \hbar \omega \\ E_2 &= \frac{5}{2} \hbar \omega \\ &= \frac{a^* b}{\alpha} e^{-i\omega t} + \frac{b^* a}{\alpha} e^{i\omega t} \end{aligned}$$

a, b 为实数

$$\langle x \rangle = \frac{2ab}{\alpha} \cos \omega t$$

第四题: (30 分)

已知一群氢原子处于波函数 $\Psi(r, t) = A\psi_{100}e^{-\frac{iE_1 t}{\hbar}} + B\psi_{210}e^{\frac{iE_2 t}{\hbar}} + C\psi_{211}e^{\frac{iE_2 t}{\hbar}}$ 所表示的量子态中, 其中 A, B, C 为复数, 且有

$$\psi_{100} = \frac{1}{\sqrt{\pi a^3}} e^{-r/2a}$$

$$\psi_{210} = \dots e^{-r/2a} r \cos \theta \quad (\text{前面的系数忘了, 但是系数不重要})$$

$$\psi_{211} = \dots e^{-r/2a} r \sin \theta e^{i\phi}$$

现在想要通过一些列测量来确定氢原子的量子态, 试求:

- 1) 能量的可能测值及相应概率;
- 2) $t=0$ 时刻, L^2 和 L_z 的可能测值及其概率;
- 3) 在 t 时刻, 求 L_x 的可能测值及其概率;
- 4) 位置分量 x, y, z 的平均值;
- 5) 测定了上述力学量后, 可以完全确定氢原子的量子态了吗? 说明

$$(1) \quad \psi_{100} \text{ 态, } L_x \text{ 可测量值为 } 0, \text{ 概率为 } 1$$

$$\psi_{210} \text{ 态, } L_x \text{ 可测量值为 } 0, \pm \hbar$$

$$\text{概率为 } P_0, P_1$$

$$(1) \quad E = -\frac{e^2}{2an^2}$$

$$E_1: |A|^2$$

$$E_2: |B|^2 + |C|^2$$

$$(2) \quad t=0$$

$$\psi = A\psi_{100} + B\psi_{210} + C\psi_{211}$$

$$L^2: 0, \hbar^2$$

$$2\hbar^2, |B|^2 + |C|^2$$

$$L_z: 0, \hbar^2 + \hbar^2$$

$$\hbar, |C|^2$$

$$(4) \quad \langle x \rangle = \int \psi^*(r, t) r \cos \theta \sin \theta \psi(r, t) d\tau$$

$$\langle \hat{L}_x^2 \rangle = 2\hbar^2 P_1 = \frac{1}{2} \langle \hat{L}^2 - \hat{L}_z^2 \rangle$$

$$= \hbar^2$$

$$P_1 = \frac{1}{2}$$

$$P_0 = 0$$

ψ_{211} 态, L_x 可测值为 $0, \pm \hbar$
 $P_0 \quad P_1$

$$\langle \hat{L}_x^2 \rangle = 2\hbar^2 P_1 = \frac{1}{2} \langle \hat{L}^2 - \hat{L}_z^2 \rangle$$

$$= \frac{1}{2} \hbar^2$$

$$P_1 = \frac{1}{4}$$

$$P_0 = \frac{1}{2}$$

综上, L_x 为 0 的概率 $|A|^2 + \frac{1}{2}|C|^2$
 $\pm \hbar$ $\frac{1}{4}|B|^2 + \frac{1}{4}|C|^2$

$$\frac{d}{dt} \int \psi^* r \cos \theta \sin \varphi \psi r^2 \sin \theta dr d\theta d\varphi$$

$$= 0 \quad \langle y \rangle = \langle z \rangle$$

5) $\overline{V_L}$

第五题: (25 分)

空间中有 x 方向均匀磁场, 大小为 B , 质量 m 电量 q 自旋为 $1/2$ 的电子在其中运动, 开始时电子处于自旋向上状态 (S_z 本征值为 $\hbar/2$), 试求:

- 1) t 时刻电子的波函数;
- 2) t 时刻电子自旋 y 分量的可能测值及概率;
- 3) 电子自旋第一次反向的时间 T ;
- 4) 在 $t = 2T$ 时刻, 对电子进行测量, 发现其处于自旋 x 分量本征值为 $\hbar/2$ 的本征态, 求 $t = 3T$ 时刻电子自旋 x 分量的可能测值及其概率

$$(1) \quad \vec{B} = B \vec{i}$$

$$\hat{H} = -\vec{\mu}_s \cdot \vec{B} = -\frac{qB}{mc} \hat{S}_x = -\frac{qB\hbar}{2mc} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$E_1 = -\frac{qB\hbar}{2mc} = \omega_s \hbar, \quad |\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$E_2 = \frac{qB\hbar}{2mc} = \omega_s \hbar, \quad |\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\omega_s = \frac{qB}{2mc}$$

$$|\psi(t)\rangle = C_1 e^{-\frac{iE_1 t}{\hbar}} |\psi_1\rangle + C_2 e^{-\frac{iE_2 t}{\hbar}} |\psi_2\rangle$$

$$t=0 \text{ 时 } |\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$C_1 = \langle \psi_1 | \psi \rangle = \frac{1}{\sqrt{2}} (1 \ 1) / \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$(2) \quad \hat{S}_y = \frac{\hbar}{4}, \quad |\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$\hat{S}_y = -\frac{\hbar}{4}, \quad |\psi_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\hat{S}_y = \frac{\hbar}{4} \text{ 的概率 } |\langle \psi_1 | \psi(t) \rangle|^2 = \frac{1}{4} \left| \begin{pmatrix} 1 & i \end{pmatrix} \begin{pmatrix} \cos \omega t \\ i \sin \omega t \end{pmatrix} \right|^2$$

$$= \frac{1}{4} (1 - \sin 2\omega t)$$

$$\hat{S}_y = -\frac{\hbar}{4} \text{ 的概率 } |\langle \psi_2 | \psi(t) \rangle|^2 = \frac{1}{4} \left| \begin{pmatrix} 1 & -i \end{pmatrix} \begin{pmatrix} \cos \omega t \\ i \sin \omega t \end{pmatrix} \right|^2$$

$$= \frac{1}{4} (1 + \sin 2\omega t)$$

$$(3) \quad |\psi(t)\rangle = \begin{pmatrix} \cos \omega t \\ i \sin \omega t \end{pmatrix}$$

$$\text{自旋反向, } |\psi(t)\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|\cos \omega t|^2 = 0$$

$$|\sin \omega t|^2 = 1$$

$$\omega_0 T = (n+1) \frac{\pi}{2}$$

$$\frac{qB}{2mc} T = (n+1) \frac{\pi}{2}$$

$$T = \frac{(n+1) \pi mc}{qB}$$

$$n=0 \text{ 时, } T = \frac{\pi mc}{qB} \text{ 时, 第一次反向}$$

$$(4) \quad t = 2T \text{ 时}$$

$$\psi(x, 2T) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|\cos 2\omega T|^2 = \frac{1}{2}$$

$$|\sin 2\omega T|^2 = \frac{1}{2}$$

$$2\omega T = \frac{\pi}{4}$$

$$T = \frac{\pi}{8\omega}$$

$$\psi(x, 3T) = \begin{pmatrix} \cos \frac{3}{8}\pi \\ i \sin \frac{3}{8}\pi \end{pmatrix}$$

$$c_2 = \langle \psi_2 | \psi \rangle = \frac{1}{\sqrt{2}}$$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} e^{i\omega t} |\psi_1\rangle + \frac{1}{\sqrt{2}} e^{-i\omega t} |\psi_2\rangle$$

$$= \frac{1}{2} \begin{pmatrix} e^{i\omega t} + e^{-i\omega t} \\ e^{i\omega t} - e^{-i\omega t} \end{pmatrix}$$

$$= \begin{pmatrix} \cos \omega t \\ i \sin \omega t \end{pmatrix}$$

$$S_x = \frac{\hbar}{2}, \quad |\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$S_x = -\frac{\hbar}{2}, \quad |\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{aligned} S_x = \frac{\hbar}{2} \text{ 的概率 } |\langle \psi | \psi(x, y) \rangle|^2 &= \frac{1}{2} \left| \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} \cos \frac{3}{8}\pi \\ i \sin \frac{3}{8}\pi \end{pmatrix} \right|^2 \\ &= \frac{1}{2} (\cos \frac{3}{8}\pi + i \sin \frac{3}{8}\pi) (\cos \frac{3}{8}\pi - i \sin \frac{3}{8}\pi) \\ &= \frac{1}{2} \end{aligned}$$

$$S_x = -\frac{\hbar}{2} \text{ 的概率 也为 } \frac{1}{2}$$

第六题: (20 分)

一维无限深势阱两端坐标为 0 和 a, 有两个全同粒子在其中运动, 试

求:

- 1) 若粒子自旋为 0, 求体系基态和第一激发态的能量和波函数;
- 2) 若粒子自旋为 1/2, 求体系基态和第一激发态的能量和波函数;
- 3) 粒子自旋为 0, 求在微扰势 $V(x) = -aV_0\delta(x_2 - x_1)$ 作用下, 体

系基态和第一激发态能量一级修正

(1) 自旋为 0 的粒子 单个粒子 $\psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \quad 0 < x < a, \quad n = 1, 2, 3, \dots$
 $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$

基态 $E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$

$\psi_I = \psi_1(x_1) \psi_1(x_2)$

第一激发态 $E_2 = \frac{5\pi^2 \hbar^2}{2ma^2}$

$\psi_{II} = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)]$

(2) 自旋 1/2, 费米子

基态 $E_I = \frac{\pi^2 \hbar^2}{2ma^2}$

$\psi_I = \psi_1(x_1) \psi_1(x_2) |00\rangle$

第一激发态 $E_{II} = \frac{5\pi^2 \hbar^2}{2ma^2}$

$\psi_{II} = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)] |00\rangle$

$$\psi_{II}' = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) - \psi_1(x_2) \psi_2(x_1)] \begin{cases} |11\rangle \\ |10\rangle \\ |1-1\rangle \end{cases}$$

$$(3) \quad V(x) = -aV_0\delta(x_2 - x_1) = \hat{V}$$

$$\text{基态 } E_1 = \frac{\pi^2 \hbar^2}{ma^2}$$

$$\psi_I = \psi_1(x_1) \psi_1(x_2) = \frac{2}{a} \sin \frac{\pi x_1}{a} \sin \frac{\pi x_2}{a}$$

$$E_1^{(1)} = H'_{11} = \int \psi_1^*(x_1) \psi_1^*(x_2) [-aV_0\delta(x_2 - x_1)] \psi_1(x_1) \psi_1(x_2) dx$$

$$= -\frac{4V_0}{a} \sin^4 \frac{\pi x}{a}$$

$$\text{第一激发态 } E_2 = \frac{5\pi^2 \hbar^2}{2ma^2}$$

$$\psi_{II} = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_1(x_2) \psi_2(x_1)]$$

$$E_2^{(1)} = H'_{22} = \frac{-aV_0}{2} \int [\psi_1^*(x_1) \psi_2^*(x_2) + \psi_1^*(x_2) \psi_2^*(x_1)] \cdot \delta(x_2 - x_1) dx$$

$$= -\frac{aV_0}{2} \int \left(|\psi_1(x_1) \psi_2(x_2)|^2 + \psi_1^*(x_1) \psi_2^*(x_2) \psi_2(x_1) \psi_1(x_2) \right. \\ \left. + |\psi_2(x_1) \psi_1(x_2)|^2 + \psi_1^*(x_2) \psi_2^*(x_1) \psi_1(x_1) \psi_2(x_2) \right) \delta(x_2 - x_1) dx$$

$$= -\frac{2V_0}{a} \int \left[\left(\sin \frac{\pi x_1}{a} \sin \frac{2\pi x_1}{a} \right)^2 + 2 \sin \frac{\pi x_1}{a} \sin \frac{2\pi x_1}{a} \sin \frac{2\pi x_1}{a} \sin \frac{\pi x_1}{a} \delta(x_2 - x_1) \right. \\ \left. + \left(\sin \frac{2\pi x_1}{a} \sin \frac{\pi x_1}{a} \right)^2 \right] \delta(x_2 - x_1) dx$$

$$= -\frac{2V_0}{a} \cdot 4 \left(\sin \frac{\pi x}{a} \sin \frac{2\pi x}{a} \right)^2$$